

DIGITAL FORENSIC EXAMINATION REPORT

National Technical Research Organisation (NTRO) — Case Reference: CASE-2026-001

Case Number:	CASE-2026-001	Target Host:	WS-FIN-04.fin.local
Case Title:	Suspicious Execution & Exfiltration on Workstation WS-FIN-04	Status:	ACTIVE
Lead Investigator:	Lead DFIR Analyst (NTRO-INV-42)	Date & Time:	2026-09-26 10:50 UTC
Forensic Framework:	NTRO DSL-DFIR v2.4 (Authorized Mode)	Evidence Hash Protocol:	FIPS 180-4 SHA-256

1. EXECUTIVE SUMMARY

An authorized forensic investigation was commenced for target host **WS-FIN-04.fin.local** under NTRO digital evidence procedures. Forensic analysis utilized non-intrusive domain-specific scripting language (Forensic DSL) commands operating under **FORENSIC_READ_ONLY** policies. A total of **5** evidence items were acquired with verified SHA-256 cryptographic digests. The deterministic rule engine identified **8** notable security findings.

2. KEY SUSPICIOUS FINDINGS

Severity	Rule & Title	Category	Evidence Reference
CRITICAL	IOC Hash Match: CobaltStrike Beacon Synthetic Stager File 'update_svc_helper.exe' (C:\Users\jdoe\AppData\Local\Temp\update_svc_helper.exe) matc...	IOC	C:\Users\jdoe\AppData\Local\Temp\update_svc_helper.exe
HIGH	Executable in Monitored User Directory: update_svc_helper.exe Executable binary or script discovered residing in high-risk directory 'C:\Users\jdoe\AppData...	FILE	C:\Users\jdoe\AppData\Local\Temp\update_svc_helper.exe
HIGH	Executable in Monitored User Directory: payload_stager.ps1 Executable binary or script discovered residing in high-risk directory 'C:\Users\jdoe\Down...	FILE	C:\Users\jdoe\Downloads\payload_stager.ps1
HIGH	Suspicious Process Execution: update_svc_helper.exe (PID:4820) Process 'update_svc_helper.exe' running from temporary path 'C:\Users\jdoe\AppData\Local\T...	PROCESS	PID-4820

HIGH	Obfuscated PowerShell Invocation Detected (PID:2104) PowerShell process invoked with evasion/download flags: powershell.exe -NonI -W Hidden -Ex...	PROCESS	PID-2104
CRITICAL	IOC Outbound Network C2 Match: 198.51.100.45:4444 Active socket initiated by update_svc_helper.exe (PID:4820) connects to known Threat Intel...	NETWORK	192.168.1.105:51442 -> 198.51.100.45:4444
CRITICAL	IOC Outbound Network C2 Match: 203.0.113.88:8080 Active socket initiated by powershell.exe (PID:2104) connects to known Threat Intel IP 203...	NETWORK	192.168.1.105:52110 -> 203.0.113.88:8080
MEDIUM	Authentication Failure Anomaly (1 event(s) recorded) Detected authentication failure events. Detailed probe against workstation credentials....	LOG	Security Log Event 4625

3. EVIDENCE INVENTORY & SHA-256 HASH VERIFICATION

ID	Artifact Name	Type	SHA-256 Digest	Integrity
EVD-CASE-202-001	System_Config_WS-FIN-04.fin.local.json	SYSTEM	e3b0c44298fc1c149afb4c8996f...	VERIFIED
EVD-CASE-202-002	update_svc_helper.exe	FILE	8f4e2a6d5c1b9e0f3d7a8b4c2e6f...	VERIFIED
EVD-CASE-202-003	payload_stager.ps1	FILE	c8a49c2d1b5e3f7a9d0b2c4e6f8a...	VERIFIED
EVD-CASE-202-004	quarterly_budget_2026.xlsx	FILE	1a2b3c4d5e6f7a8b9c0d1e2f3a4b...	VERIFIED
EVD-CASE-202-005	invoice_q3_confidential.pdf	FILE	7d8e9f0a1b2c3d4e5f6a7b8c9d0e...	VERIFIED

4. CHAIN OF CUSTODY AUDIT TRAIL

Timestamp	Action	Investigator	Details & Chain Digest
2026-09-26T10:50:22	EVIDENCE_ACQUIRE D	Lead DFIR Analyst	Acquired update_svc_helper.exe (FILE). Initial SHA-256 calculated: 8f4... Digest: d93197654f6f9080b161...
2026-09-26T10:50:22	EVIDENCE_ACQUIRE D	Lead DFIR Analyst	Acquired payload_stager.ps1 (FILE). Initial SHA-256 calculated: c8a49c... Digest: e62a8aa6650016aa29c7...
2026-09-26T10:50:22	EVIDENCE_ACQUIRE D	Lead DFIR Analyst	Acquired quarterly_budget_2026.xlsx (FILE). Initial SHA-256 calculated... Digest: 98fa9d869b2dbd676b31...

2026-09-26T10:50:22	EVIDENCE_ACQUIRE D	Lead DFIR Analyst	Acquired invoice_q3_confidential.pdf (FILE). Initial SHA-256 calculate... Digest: 52830ecdec357bbaca6a...
2026-09-26T10:50:23	TIMELINE_CORRELA TED	Lead DFIR Analyst	Reconstructed and correlated unified chronological timeline comprising... Digest: 4b658abf02e52424e79e...
2026-09-26T10:50:23	RULE_ANALYSIS_CO MPLETED	Lead DFIR Analyst	Deterministic rule engine evaluation completed: 8 suspicious findings ... Digest: 9baa93e1985995018580...

5. INVESTIGATOR ATTESTATION & CONCLUSION

I hereby attest that the digital evidence items cataloged in this report were acquired and processed in accordance with standardized forensic procedures. All operations executed strictly within read-only boundaries under authorized cryptographic control. No system state was illegally modified.

Examiner: Lead DFIR Analyst

Badge: NTRO-INV-42

Signature: [CRYPTOGRAPHICALLY SEALED BY
NTRO-INV-42]

Date: 2026-09-26 10:50:23 UTC